

DSK817 - Regression Diagnostics

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```
library(car)
library(glmnet)
library(knitr)
library(kableExtra)

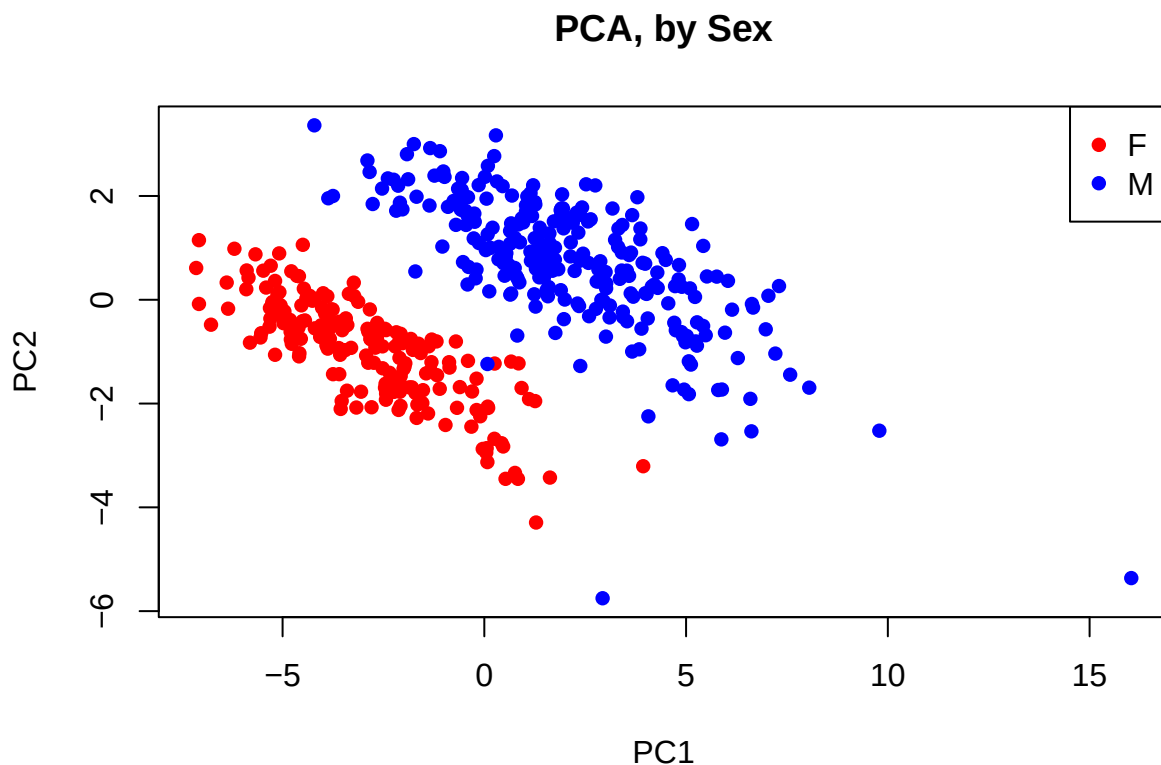
setwd('/Users/fangsiyu/Desktop/sdu-2026-code/817_statistic_modeling/project')
df <- read.csv("data/examdata.csv")
df$height <- df$height * 100

# =====
# STEP 1.1: PCA 計算與繪圖
# =====
pca_res <- prcomp(df[, -c(1, 2)], center = TRUE, scale. = TRUE)
pc1 <- pca_res$x[, 1]
pc2 <- pca_res$x[, 2]

point_colors <- ifelse(df$sex == "F", "red", "blue")

# 繪圖 (這會直接出現在 PDF 裡)
plot(pc1, pc2,
     col = point_colors,
     pch = 16,
     xlab = "PC1",
     ylab = "PC2",
     main = "PCA, by Sex")

legend("topright",
     legend = c("F", "M"),
     col = c("red", "blue"),
     pch = 16)
```



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# =====
# STEP 1.2: 使用 kable 產出 PCA 摘要表格
# =====

# 提取重要性資訊
pca_importance <- as.data.frame(summary(pca_res)$importance)

# 產出符合 PDF 格式的表格
kable(pca_importance,
      digits = 3,
      caption = "PCA 變異數解釋比例摘要",
      align = "c") %>%
  kable_styling(latex_options = c("striped", "hold_position"),
                full_width = F) %>%
  add_header_above(c(" " = 1, "Principal Components" = ncol(pca_importance)))
```

Table 1: PCA 變異數解釋比例摘要

	Principal Components											
	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10	PC11	PC12
Standard deviation	3.423	1.434	0.958	0.742	0.640	0.522	0.485	0.442	0.369	0.333	0.322	0.282
Proportion of Variance	0.689	0.121	0.054	0.032	0.024	0.016	0.014	0.012	0.008	0.007	0.006	0.005
Cumulative Proportion	0.689	0.810	0.864	0.896	0.921	0.937	0.950	0.962	0.970	0.977	0.983	0.987

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# 直接把你的三個模型丟進去
stargazer(model_1_ols_all, model_2_robust, model_3_ols_clean,
  type = "text",          # 如果是在 Console 看，用 "text"；輸出報告建議用 "html" 或 "l
  title = "敏感度分析：模型係數比較表",
  column.labels = c("包含39(OLS)", "包含39(Robust)", "剔除39(OLS)"),
  digits = 3,             # 小數點位數
  model.names = FALSE,    # 隱藏預設的模型名稱
  dep.var.labels = "體脂率 (%)", # 設定應變數標籤
  covariate.labels = c("身高", "腹圍", "年齡", "手腕", "截距")) # 設定變數名稱

```